QQ-C-585 B
November 15, 1974
SUPERSEDING
Fed. Spec. QQ-C-585 A
December 31, 1963

FEDERAL SPECIFICATION

COPPER-NICKEL-ZINC ALLOY PLATE, SHEET, STRIP, AND BAR (COPPER ALLOY NUMBERS 735, 745, 752, 762, 766 and 770)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers six alloys of copper-nickel-zinc flat products with slit, slit and edge-rolled, sheared, sawed, or machined edges (plate, sheet, strip, and rolled bar) used for fabricated or machined items, but does not include flat products with finished edges.

1.2 Classification.

1.2.1 Alloys, forms, and tempers. The material covered by this specification shall be furnished in the following alloys, forms, and tempers, as specified (see 6.2):

Alloys:
735
745
752
762
766
770

FSC 9530, 9535

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Forms

Plate
Sheet
Strip
Rolled bar

Tempers

Annealed:

Light (0.015 mm nominal grain size)
Soft (0.035 mm nominal grain size)
Dead soft (0.070 mm nominal grain size) (Alloys 745, 752, and 770)

Rolled:

Quarter-hard
Half-hard
Hard
Extra-hard
Spring (except alloy 735)

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issues in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Standards:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civilian Agencies).
Fed. Std. No. 146 - Tolerances for Copper and Copper Base Alloy Mill Products.
Fed. Test Method Std. No. 151 - Metals; Test Methods.
Fed. Std. No. 185 - Identification Marking of Copper and Copper Base Alloy Mill Products.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D.C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Ft. Worth, Denver, San Francisco, Los Angeles, and Seattle, Wash.

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(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specification:

MIL-C-3993 - Copper and Copper Base Alloy Mill Products, Packaging of.

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by Contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM) Standards:

E 8 - Tension Testing of Metallic Materials
E 112 - Estimating the Average Grain Size of Metals
E 290 - Semi-Guided Bend Test for Ductility of Metallic Materials

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphis, Pennsylvania 19103.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Manufacture. The material shall be manufactured by such hot or cold working, annealing, coiling, or straightening as may be necessary to meet the requirements of this specification.

3.2 The material shall be furnished in flat, straight lengths unless material in coils (rolls) or on reels or bucks is specified (see 6.2).

3.3 Chemical composition.

3.3.1 The material shall conform to the chemical requirements specified in table I.

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Table I. Chemical composition, percent

Copper Alloy Number	Copper	Nickel[1/]	Lead, max.	Iron, max.	Manganese, max.	Zinc, max.
735	70.5-73.5	16.5-19.5	0.10	0.25	0.50	Rem.
745	63.5-66.5	9.0-11.0	.10	.25	.50	Rem.
752	63.0-66.5	16.5-19.5	.10	.25	.50	Rem.
762	57.0-61.0	11.0-13.5	.10	.25	.50	Rem.
766	55.0-58.0	11.0-13.5	.10	.25	.50	Rem.
770	53.5-56.5	16.5-19.5	.10	.25	.50	Rem.

1/ Cobalt counted as nickel.

3.3.2 These specification limits do not preclude the possible presence of other unnamed elements. However, analysis shall regularly be made only for the minor elements listed in the table, plus either copper or zinc or plus all major elements except one. The major element which is not analyzed shall be determined by difference between the sum of those elements analyzed and 100 percent. By agreement between the manufacturer and the procuring agency, analysis may be required and limits established for elements not specified.

3.4 Mechanical properties.

3.4.1 Tensile strength. Material in rolled tempers shall conform to the tensile strength specified in table II.

Table II. Tensile strength of rolled temper material, ksi.

Copper Alloy Number	Temper				
	Quarter-hard	Half-hard	Hard	Extra-hard	Spring
735	56-69	63-75	73- 84	79- 90	--
745	56-73	67-82	80- 94	89-102	95-108
752	58-72	66-80	78- 91	86- 98	90-101
762	65-81	75-91	90-105	99-114	107-122
766	65-81	75-91	90-105	101-114	109-122
770	69-87	78-95	92-109	102-117	108-123

3.4.2 Bend properties.

3.4.2.1 Rolled tempers. When specified (see 6.2), material in the quarter-hard and half-hard tempers shall withstand, without cracking, being bent at room temperature through an angle of 90 degrees across the direction of rolling around a radius equal to 1/2 the thickness of the plate, sheet, strip, or bar.

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3.4.2.2 Annealed tempers. When specified (see 6.2), material in the annealed tempers shall withstand, without cracking, being bent at room temperature flat on itself across the direction of rolling.

3.5 Grain size of annealed tempers. Material in the annealed tempers shall conform to the grain-size requirements shown in table III.

Table III. Permissible variation in grain size for annealed tempers.

Annealed temper	Nominal grain size, mm, min.	Permissible variation in average grain size, mm.	
		Min.	Max.
Light	0.015	1/	0.025
Soft	.035	0.025	.050
Dead soft	.070	.050	.100

1/ Although no minimum grain size is specified, the material shall be fully recrystallized.

3.6 Dimensional tolerances. The following references of Fed. Std. No. 146 shall apply:

Dimension	Reference
Thickness	1b (1)
Width	1b (2)
Length	1b (3), 1b (4)
Straightness	1b (5)

3.7 Identification marking. When specified (see 6.2), product identification marking shall be in accordance with Fed. Std. No. 185 (see 6.2 and 6.3).

3.8 Workmanship. Material shall be uniform in quality and temper, clean, sound, smooth, and free from foreign material, pipes, slivers, laps, cracks, seams, scale, burrs, buckles, damaged ends or edges, and other defects which, due to their nature, degree, or extent, detrimentally affect the serviceability for the intended parts. Material shall be commercially straight or flat unless coils or rolls are specifically specified (see 6.2). When some characteristic of workmanship is especially important to the serviceability of the intended parts, this information shall be included along with the character of the application of material in the ordering data (see 6.2).

4. QUALITY ASSURANCE PROVISIONS

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4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Lot. Unless otherwise specified (see 6.2), a lot shall consist of material of the same alloy, form, temper, and size, submitted for inspection at one time consisting of:

a. 10,000 lbs. or fraction thereof or

b. The mixture of two or more furnace charges or crucible melts weighing not more than 70,000 lbs. into a single ladle or holding furnace used to pour one or more castings at the same time.

4.3 Sampling. Samples for the purpose of tests prescribed in this specification shall be selected in a manner as to correctly represent the material furnished and avoid needless destruction of finished material when samples representative of the material are available from other sources.

4.3.1 For chemical analysis. The number of samples specified in table IV shall be selected from different pieces in each lot. The selected pieces shall be used to prepare samples for analysis in accordance with method 111 or 112 of Fed. Test Method Std. No. 151.

Table IV. Sampling for chemical analysis

Pounds of material in lot	Number of samples[1/]
Up to 2500, incl.	1
Over 2500 to 5000 incl.	2
Over 5000 to 7500 incl.	3
Over 7500 to 10,000 incl.	4
Over 10,000 to 70,000 incl.	1[2/]

1/ If the number of original bars, billets, or cakes from which the material is processed is less than the number of samples specified, not more than one sample need be taken from each piece.

2/ See 4.2b.

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4.3.2 For mechanical properties.

4.3.2.1 For tension test. Unless otherwise specified (see 6.2), two tension test specimens shall be taken from each lot, and each shall be selected from a different piece unless the lot consists of one piece in which case one test specimen shall be sufficient.

4.3.2.2 For bend test. When specified (see 6.2), two bend test specimens shall be taken from each lot and each shall be selected from a different piece.

4.3.3 For grain size. Two pieces from each lot shall be selected for grain size determinations. Pieces selected for visual or dimensional examination or the ends of bend test samples may be used.

4.3.4 For visual and dimensional examination.

4.3.4.1 Pieces weighing over 150 pounds. Each piece shall be examined.

4.3.4.2 Pieces weighing 150 pounds or less. From each lot of material with pieces weighing 150 pounds or less, a representative sample shall be selected in accordance with MIL-STD-105, inspection level II, with an acceptable quality level (AQL) of 1.5 percent defective. The samples selected for dimensional examination may be the same as those selected for visual examination, but shall be evaluated separately.

4.3.4.3 When material is furnished in coils (rolls) or on reels, or bucks, the sample for examination shall be taken from within 10 feet of the outer end. If the sample selected is rejected due to handling marks, an additional 20 feet shall be used for re-examination.

4.4 Examination.

4.4.1 Visual and dimensional. Pieces selected in accordance with 4.3.4 shall be visually examined to determine compliance with the requirements for identification marking (see 3.8) and workmanship (see 3.9) and shall be measured for compliance with the dimensional requirements (see 3.7).

4.4.2 Straightness. Straightness shall be determined by placing the piece or sample unit on a level surface so that the arc or departure from straightness is horizontal. The maximum depth of arc shall be measured to the nearest 1/32 inch by means of a straightedge and a steel scale.

4.4.3 Preparation for shipment. Examination of the packing and marking for shipment shall be made for conformance to the requirements of section 5.

4.5 Tension test specimens. Tension test specimens for sheet and strip, and for bar and plate up to 3/8 inch, inclusive, in thickness shall be machined to the form and dimensions of the standard rectangular tension test specimen of ASTM E 8. Tension test specimens for plate and bar over 3/8

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inch thick shall be machined to the form and dimensions of the standard rectangular or the standard round tension test specimen of ASTM E 8, or tested in full section. The longitudinal axis of the specimen shall be parallel to the direction of rolling. For material up to and including 1-1/2 inches in thickness, the central axis of the test specimen shall coincide with the central axis of the material. For material over 1-1/2 inches in thickness, the central axis of the specimen shall be located midway between the center and surface of the piece.

4.6 Test procedures.

4.6.1 Chemical analysis. The samples selected in accordance with 4.3.1 shall be analyzed in accordance with method 111 or 112 of Fed. Test Method Std. No. 151 to determine conformance with 3.3. A single analysis of the composite sample may be made. In case of dispute, analysis by method 111 shall be the basis for acceptance.

4.6.2 Tension tests. Specimens from samples selected in accordance with 4.3.2 shall be tested in accordance with ASTM E 8.

4.6.3 Bend tests. Bend tests shall be conducted in accordance with ASTM E 290.

4.6.4 Grain size requirements. Grain size shall be determined in accordance with ASTM E 112.

4.7 Rejection.

4.7.1 Examination defects. Any sample unit having one or more defects shall be rejected. If the number of nonconforming sample units in the sample exceeds the acceptance number specified in 4.3.4.2 for that sample size, the entire lot shall be rejected subject to the provisions of the section on "Acceptance and Rejection" of MIL-STD-105.

4.7.2 Test Failures. A lot shall be rejected for failure to meet any of the test requirements when tested in accordance with 4.6, subject to the provisions of the section on "Rejection and Retests" of Fed Test Method Std. No. 151.

5. PREPARATION FOR DELIVERY

5.1 Packing (see 6.2 and 6.3).

5.1.1 Levels A and B. The material shall be packed in accordance with MIL-C-3993.

5.1.2 Level C. The products shall be separated by size, alloy, form, and temper and packed in accordance with the manufacturer's standard practice into containers of a type and size commonly used for the purpose, in such a

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manner as to insure acceptance by carrier for transportation at the lowest rate applicable and to afford maximum protection from normal hazards of transportation.

5.2 Marking (see 6.2 and 6.3).

5.2.1 Civil agencies. In addition to markings required by the contract or order, shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.2.2 Military agencies. In addition to markings required by the contract or order, shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. This specification covers six alloys of copper-nickel-zinc alloy suitable for use as hardware and optical goods where color and corrosion resistance are factors in the application. The alloys containing high percentages of zinc are suitable for springs.

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

a. Title, number, and date of this specification.
b. Alloy, form, and temper required (see 1.2).
c. Size of material.
d. When material is to be furnished in coils (rolls), or on reels or bucks (see 3.2 and 3.8).
e. When bend test is required for material in the quarter-hard, half-hard, or annealed temper (see 3.4.2.1, 3.4.2.2, and 4.3.2.2).
f. Lengths, whether specific or stock lengths with or without ends (see 3.6).
g. When product identification marking is required (see 3.7).
h. When some characteristic of workmanship is important, the characteristic and the application of the material (see 3.8).
i. When a lot size other than as specified in 4.2 is required.
j. When other than two tension-test specimens per lot are required (see 4.3.2.1).
k. Whether material is to be packed by level A, B, or C (see 5.1).
l. Special marking if required (see 5.2).
m. Maximum gross weight of containers.

6.3 The requirements for product identification marking (see 3.8) and for packing and marking for shipment (see 5.1 and 5.2) specified herein apply to direct shipment for Government activities and apply also, where specified, to contracts or orders between the manufacturer and the Government prime contractor.

6.4 Definitions.

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6.4.1 Bar. As covered by this specification, a solid rectangular section or one with two plane parallel surfaces and slit, sheared, sawed, or machined edges, up to and including 12 inches in width and over 0.188 inch in thickness.

6.4.2 Plate. A flat rolled product over 3/16 inch (0.188 inch) in thickness and over 12 inches in width.

6.4.3 Sheet. A flat rolled product up to and including 0.188 inch in thickness and over 24 inches in width.

6.4.4 Strip. As covered by this specification, a flat product, other than flat wire, up to and including 0.188 inch in thickness, and furnished with slit, sheared, or slit and edge rolled edges in widths up to 24 inches, inclusive.

6.5 General information.

6.5.1 The thickness of all flat products should be stated in decimals of an inch.

6.5.2 Plate, sheet, and strip should be ordered in as narrow widths as can be used.

6.5.3 For purposes of weight calculations, the weight per cubic inch may be taken as 0.314 pound.

6.5.4 Where no description of the edge of plate, sheet, or strip is specified, the edges furnished will be the finish (slitting, shearing, sawing) most available to the producer.

6.5.5 When material is ordered in the form of plate, bar, sheet, and strip, it should be understood that these terms refer merely to the general form and dimensions of the material, and do not have any technical significance as to the method of manufacture.

6.5.6 Flat products in rolls are generally more economical than material of the same cross-section in straight or flat lengths.

6.5.7 Sheet and strip may also be furnished in longer commercial lengths in rolls or on reels or bucks. Plate or bar may be furnished in flat lengths or in rolls.

6.6 Related specification. Flat products with finished edges (flat wire, strip and bar) are covered under QQ-C-586.

6.7 The term "ksi" is defined as "thousand pounds per square inch". The term "psi" is obsolete.

6.8 Metric equivalents. To obtain tensile strength values in mega-Pascals (MPa), multiply the values in table II by 6.89.

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MILITARY CUSTODIANS:

Army - MR
Navy - AS
Air Force - 84

Review activities:

Army - MI
Navy - AS
Air Force - None
DSA - IS

User activities:

Army - ME, AT, EL
Navy - SA, SH
Air Force - None

Preparing activity:

Army - MR

CIVIL AGENCY COORDINATING ACTIVITY:

GSA-FSS

Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See section 2 of this specification to obtain copies and other documents referenced herein. Price 35 cents each.

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SPECIFICATION ANALYSIS SHEET

Form Approved Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION	Copper-Nickel-Zinc Alloy Plate, Sheet, Strip, and Bar
QQ-C-585B	(Copper Alloy Numbers 735, 745, 752, 762, 766 and 770)

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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